

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables that lets business teams answer their own data questions; sub-agent orchestration with human-in-the-loop validation, 8 deterministic SQL rules, and a 23-case golden-query evaluation suite enforcing accuracy before release. Approved by Sr. Director and product leadership; the only AI skill in active use by business teams on Data Ventures.
- Built and deployed a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; decision logs and state persist to BigQuery for full auditability. Designed a 3-tier event-driven v2 platform self-configuring via a BigQuery event registry — new key metrics and thresholds ship without code deploys.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users), slicing metrics across multiple dimensions; cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.
- Built an autonomous Jira ticket-resolution agent that leverages AI to enhance routine analytics: 6-gate triage, MCP context gathering, RAG over 11,000 historical SQL queries, self-healing execution. Closed 400+ data requests across 8 categories over 15 months, cutting turnaround from 30–60 minutes to under 10 minutes each.
- Replaced ~20 recurring recruitment workflows with end-to-end automation on a 68-table BigQuery platform; stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates, with data-integrity validation built in. A recurring 3-hour task now takes ~3 minutes of setup.
- Partnered with Product, Data Science, and business stakeholders to translate ambiguous questions into precise specs; presented insights and agent outputs to both technical and non-technical audiences.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Analytics & BI: SQL, Python, BigQuery, dbt, statistical modeling (logistic regression), key-metric / threshold design, data-integrity validation, root-cause analysis

Visualization & reporting: Power BI (DAX, XMLA TMSL), Tableau, LookML / Looker, Streamlit, Matplotlib, React Recharts

AI-forward: prompt engineering, evaluation suites (golden sets, human-in-the-loop validation), RAG, multi-model routing, headless monitoring agents, decision logging / observability

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

APIs / triggers / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, JSONL state stores

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL